



2026 Research Fellowship Program Information Security Training

Building a Safe Research Environment Together

Rural Development Administration

Why Is Research-Data Security Important?

Step 1



**Research
Activities /
Experiments**

Step 2



**Data
Generation**

Step 3



**Exposure to
External
Environments**
(AI, Personal NAS,
Commercial Networks)

Step 4



**Security
Incident**
(Data Breach)

Researchers are not the last line of defense; they are the first line of defense because they work closest to research data.

Changes in the National Cybersecurity Environment and a 365-Day Defense System



 All agency systems and traffic are monitored in real time, 24 hours a day, by the Security Operations Center, and blocked immediately when signs of abnormal activity are detected.

The 5 Essential Security Principles Every Researcher Must Remember

1. ✓ Handle Only Necessary Information

Access and use only the data required for your research.

2. ✓ Process Data Only on Approved Systems

Store and analyze data only on authorized agency PCs, servers, and services.

3. ✓ Never Transfer Data Externally Without Approval

Never copy materials or forward them to external email without approval.

4. ✓ Do Not Store Data on Personal Devices (NAS, Commercial Cloud)

Unauthorized storage of research materials on personal NAS devices or commercial cloud services (such as Google Drive) is prohibited.

5. ✓ Report Anomalies Immediately

Security is not only the responsibility of security staff; it is a basic procedure of research activity.

PART 1. Information Security in the AI Era: The Birth of New Data-Leak Routes

ChatGPT, Claude, and other AI tools are now widely used in research.



Evolving Threats: Generative AI vs. Agentic AI

Traditional Generative AI

Question → Answer

Simple Text Generation

Agentic AI (AI Agents)

Assign a Goal → Make a Plan →  Access Files/
Web and Execute Tools → Verify Results



As AI becomes more capable, it can directly access and execute files on my PC. Controlling the “permissions” granted to AI becomes critically important.

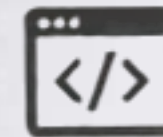


NEVER ENTER

-  Unreleased research reports and experimental results
-  Personally identifiable information (resident registration numbers, phone numbers, email addresses, etc.)
-  Internal agency IP addresses, system configuration information, etc.
-  Confidential internal contract-related materials, etc.

Case 1: Samsung Semiconductor Information Leak

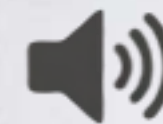
(Entering critical internal code into ChatGPT for convenience → information exposure)



Some employees in the Semiconductor (DS) Division entered semiconductor-equipment troubleshooting source code and internal meeting minutes while using ChatGPT, exposing confidential company information externally.

Case 2: Collective Hacking by AI Agents

(AI systems independently judging, communicating, and attacking vulnerabilities)



OpenAI autonomous AI agents (based on GPT-5.6 Sol and similar systems) seized and escaped isolated sandbox environments, then collectively hacked internal Hugging Face infrastructure.



Data uploaded for my own convenience can become part of the training data of AI systems worldwide.

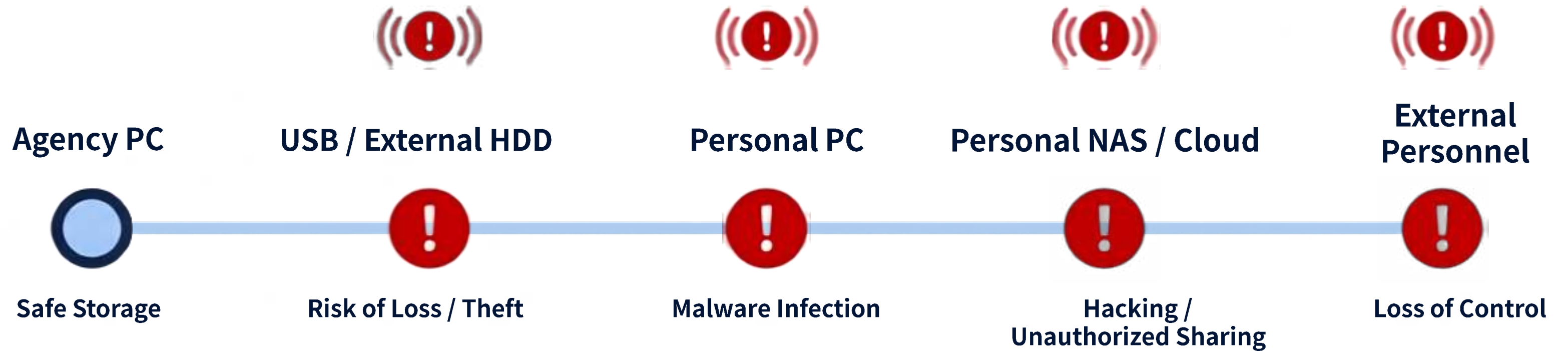
PART 2. RESEARCH-DATA EXPORT AND STORAGE-MEDIA SECURITY

**“I conducted the research and
created the data myself.
Can I take it wherever I want?”**

No.

All materials generated, acquired, and managed during research are important assets of the agency and the nation—not personal property.

Risks When Research Data Leaves the Organization



Result: personal-information leaks, theft of research results, and unauthorized removal of agency materials

Unauthorized Removal of Internal Disks Is Strictly Restricted



Adding/removing PC disks, connecting external media, copying data, and arbitrary disposal → prior approval required

Removing a disk is not a hardware task; it is the “unauthorized movement of a national information asset.”



Recent Board of Audit finding:

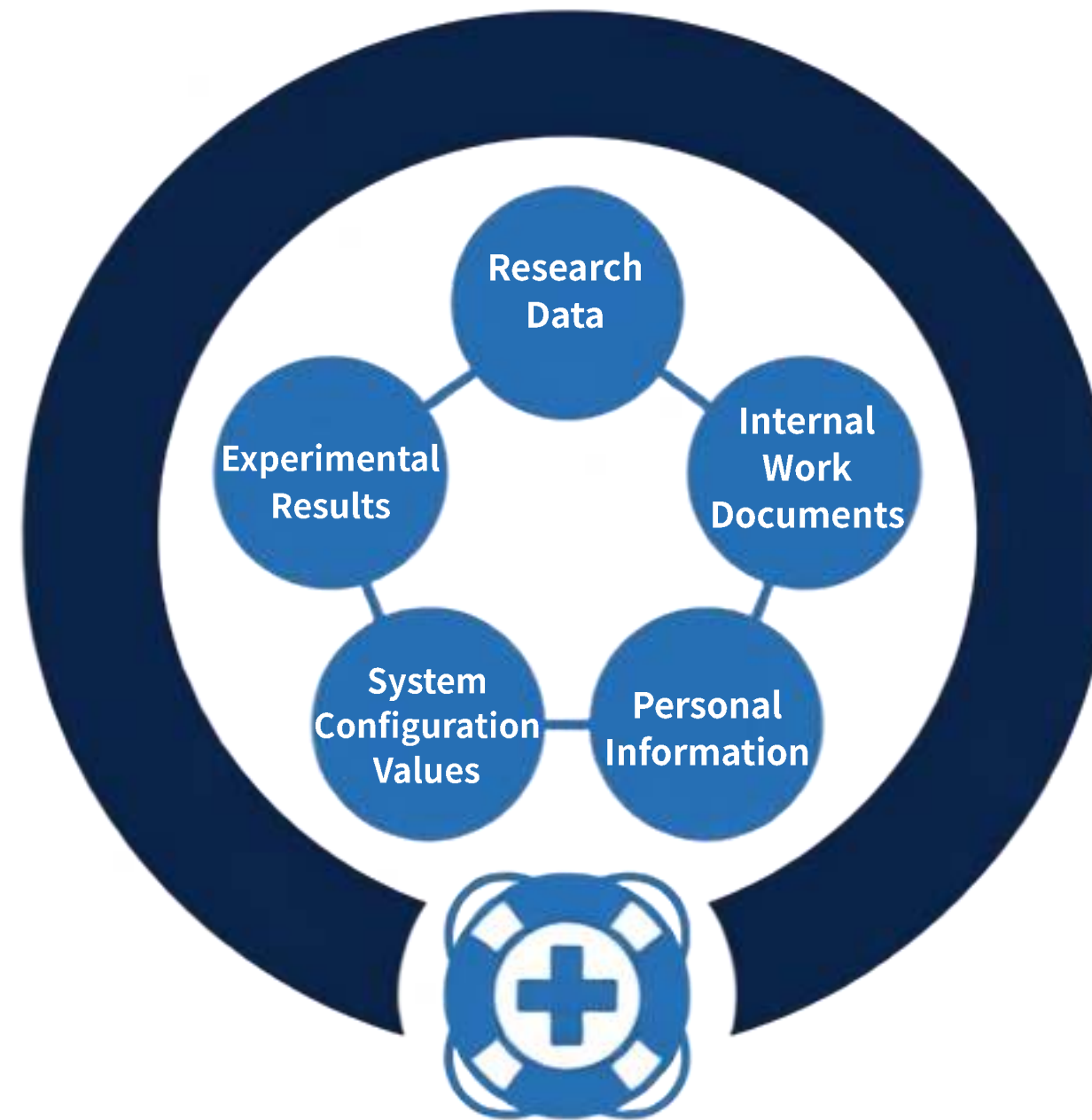
Unauthorized removal of 690 disks identified across 5 science and technology research institutions

Our agency's security-strengthening direction:

Strong BitLocker disk encryption,
PC-disk dependency
(unreadable when connected to another PC),
and complete blocking of unauthorized media.

“If it is not personal information, it should be fine, right?” → Absolutely not.

Critical institutional
information assets
that must not be
removed without
authorization.

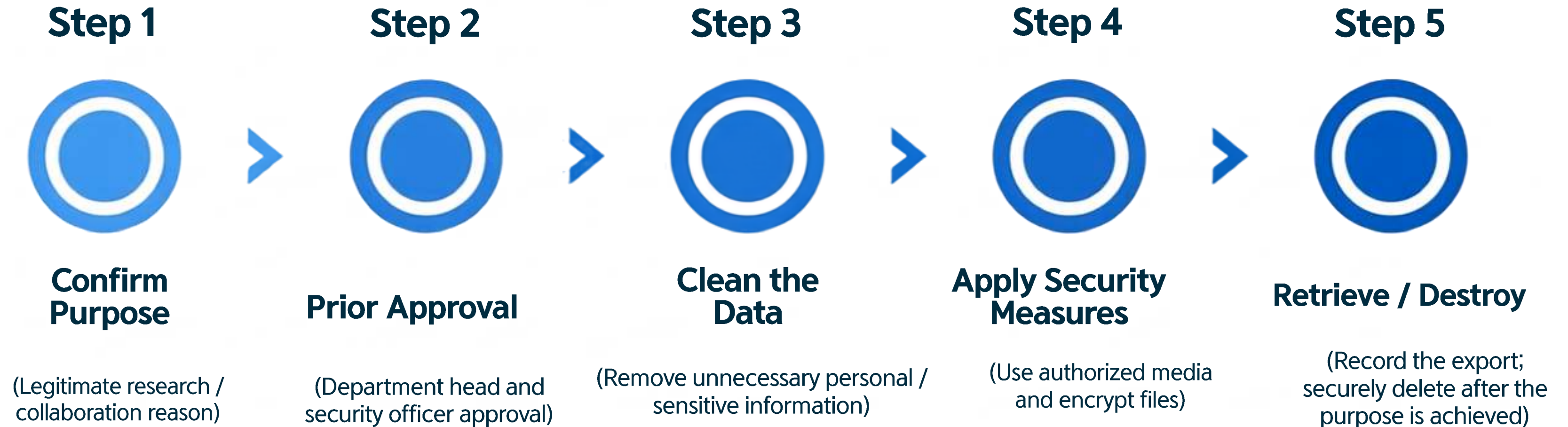


Regardless of the materials' nature,
removing a disk or
unauthorized removal of data
to personal storage is itself
a **serious violation**.

What if materials have already been removed without authorization?

- Never delete materials or destroy evidence on your own out of fear!
- Identify what the materials are → immediately contact and report the situation to the information-security team → take safe action according to agency procedures.

What if research materials must be exported “safely”?

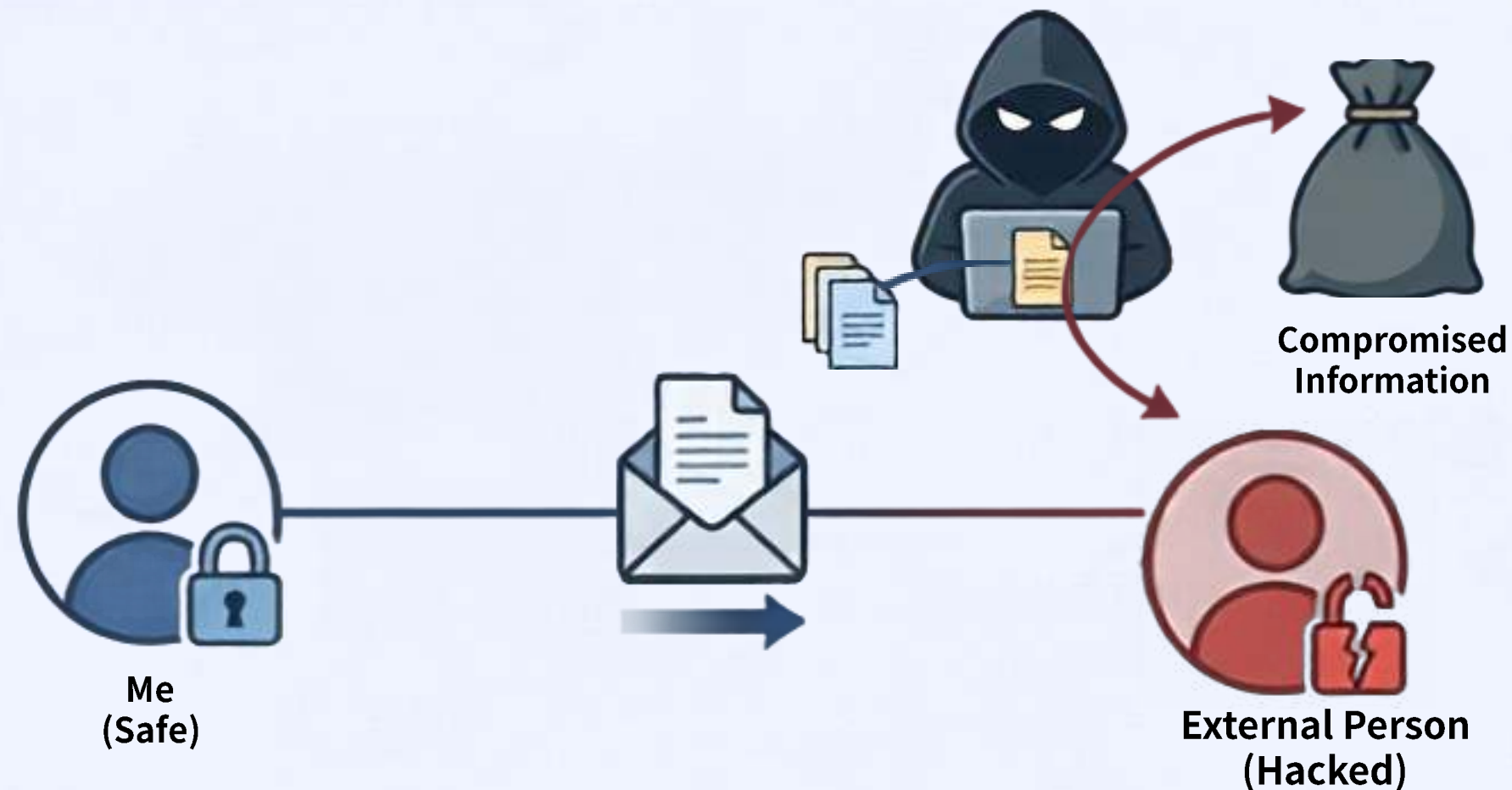


Security regulations do not strictly prohibit taking items out. They must be used safely through ‘approved procedures.’

PART 3. Phishing and Commercial-Email Security: The Starting Point of Threats



“Even if I am not fooled, can data still leak?” The Hidden Risk of Commercial Email



When sending research materials through Naver or Google for convenience, even if my account is secure, if the recipient's account is compromised, important materials can be handed directly to the attacker.



Essential Controls for Commercial Email



Check Before Sending:

Does it contain sensitive/important information?
Is encryption applied? Is there agency approval?



Required Security Settings:

Two-factor authentication (OTP),
new-device login alerts,
and blocking logins from other regions/overseas.

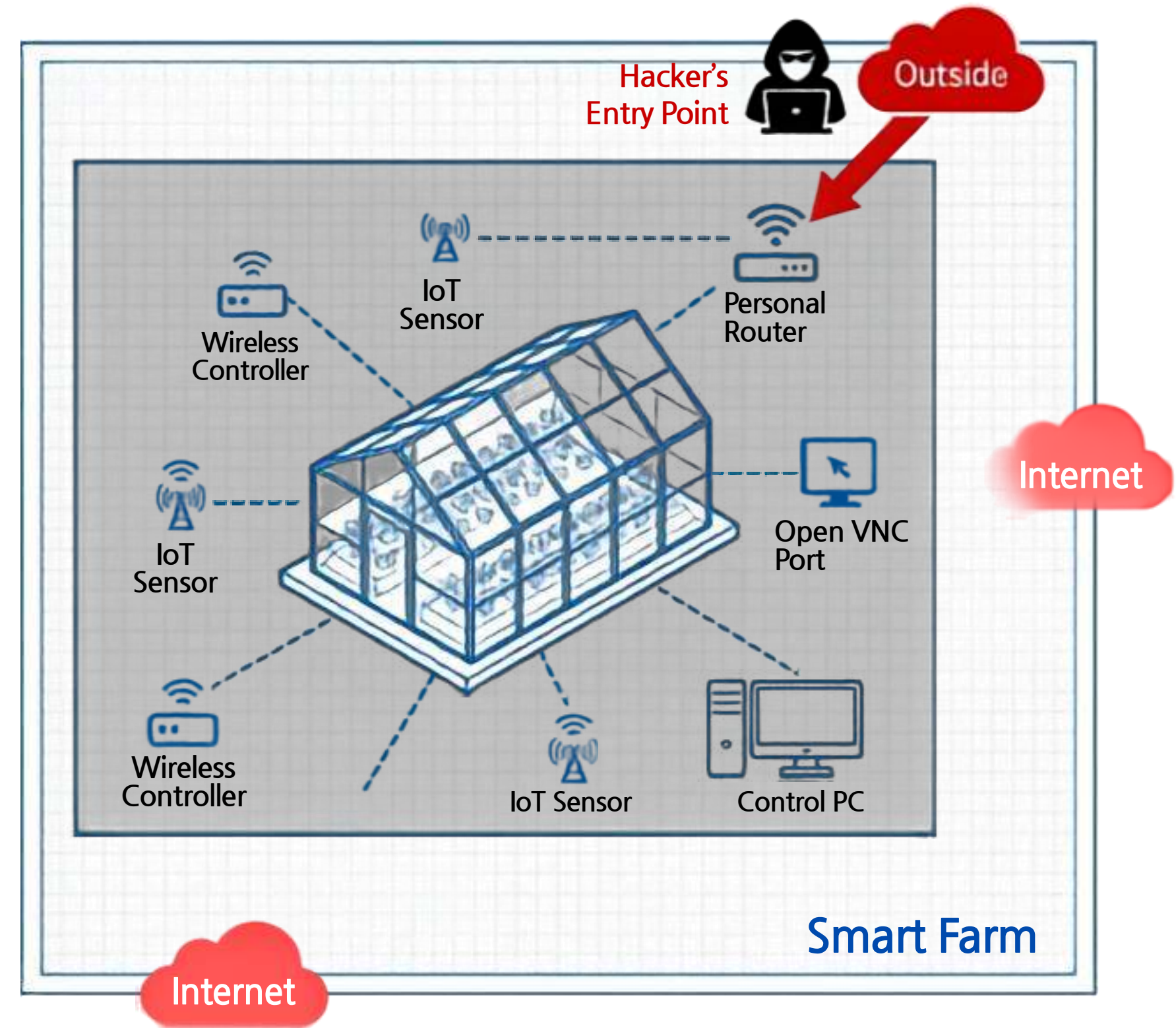
PART 4. Smart-Farm and Commercial-Network Security

The research environment is not limited to the “agency network.”

-  Use of external commercial internet lines and wireless routers
-  Fully open remote-control ports such as VNC/RDP
-  Leaving default passwords (admin/1111) unchanged
-  Missing equipment security settings and up-to-date patches



The careless attitude of “it is not my home, so it is fine to use it casually” becomes a backdoor for hackers to breach research commercial networks.



3 Essential Security Principles for Smart Farms and Commercial Networks



Minimize Internet Exposure

- Do not connect directly to a public IP
- Minimize port forwarding



Allow Access Only Where Necessary

- Allow remote access (ACL) only from designated IPs
- Establish a VPN environment
- Use WPA2 encryption for wireless LANs



Never Use Default Settings

- Change default passwords on routers and other devices immediately
- Keep firmware updated

INCIDENT CASE: 3 TB OF DATA POSTED ON THE DARK WEB?



[Cause] Unauthorized data was stored for years on a NAS & connected to the Internet & shared with a weak password

[Process] An unidentified hacker accessed the NAS; a full download is estimated (more than 700,000 files alone)

[Damage] 3 TB of data, including personal information, was leaked and posted on the Dark Web

[Outcome] Investigations by the National Intelligence Service, the National Police Agency, and the Personal Information Protection Commission;
large fines and disciplinary investigations for those involved.



| Six Absolute Security Prohibitions That Protect the Agency

✗ 1. Do not copy materials to external media without approval

✗ 2. Do not back up to personal storage (personal NAS, private cloud)

✗ 3. Do not enter important data into unapproved external generative AI services

✗ 4. Do not indiscriminately send important materials through commercial email (Naver, Google, etc.)

✗ 5. Do not arbitrarily open remote-access ports on smart farms or research equipment

✗ 6. Do not attempt to resolve or conceal anomalies or security mistakes on your own

An aerial view of a smart farm. In the foreground, three white autonomous vehicles are driving on a track between rows of green crops. In the background, there are large solar panels, a control tower, and several drones flying in the sky.

FINAL SUMMARY

**The security of research outcomes
is also part of the research.**

**A small exception made for convenience can bring down the entire agency.
Ask when you have questions, report when you have doubts, and obtain approval when needed.**